Boggadi Sudheer Kumar Reddy

Decision Sciences| Indian Institute of Management, Bangalore, Karnataka, 560076
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RESEARCH INTERESTS

Optimization Theory, Mixed-Integer Programming, Convex and Conic Optimization, Scheduling Algorithms, Networks and Graphs, Sports Analytics.

EDUCATION

Ph.D. in Decision Sciences 2021-2026*
Indian Institute of Management Bangalore, India

- **Dissertation:** Second-Order Cone Programming (SOCP)-Based Constructs For Solving Mixed-Integer Programs With Applications To Batch Scheduling Problems
- **Advisor:** Prof. Jitamitra Desai

B.Tech in Mining Engineering 2016-2020
Indian Institute of Technology, Kharagpur *Department Rank 1*

PUBLICATIONS

Conference Proceedings

1. **Reddy, B.S.K.,** & Desai, J. (2025). A Hierarchy of SOCP-based relaxations for 0-1 Quadratic Programs. *IISE Annual Conference Proceedings*, Norcross, 1-6. DOI/Link

WORKING PAPERS

1. **A Hierarchy of SOCP Relaxations for 0-1 Integer Programming Problems**
   (Co-authored with Jitamitra Desai.)
   Abstract: Several methodologies, such as the well-known *Reformulation-Linearization Technique* (RLT), produce a hierarchy of ever-tightening *linear programming* (LP) relaxations for constructing the convex hull of an underlying pure 0-1 linear program. In this research, we add to the body of literature on hierarchical approaches by employing *second-order cone programming* (SOCP)-based relaxations to construct tight relaxations for an underlying 0-1 integer program. As with other techniques, this hierarchical procedure also generates a sequence of ever-tightening SOCP relaxations that ultimately converges to the convex hull of the underlying 0-1 IP in n steps (where n is the number of binary variables in the original problem). Moreover, this methodology also offers many computational advantages as the number of constraints required at the k^{th} step of this procedure is significantly lower as compared to existing methodologies while yet retaining the strength of the underlying lower bound. Furthermore, preliminary computations on several well-known instances from QPLIB and QAPLIB are used to demonstrate the efficacy of the proposed methodology.

2. **A Continuous Time Model For Flexible Batch Scheduling With Sequence-Specific Changeovers**
   (Co-authored with Reshma Chirayil, Arvind Kumar, and Jitamitra Desai.)
   Abstract: This paper addresses the batch scheduling problem in a manufacturing industry setting, considering key factors such as flexible and variable batch sizes, multiple production lines, minimum batch-size requirements, Stock Keeping Unit (SKU) dependencies, and

delivery requirements across the scheduling horizon. A novel Mixed Integer Linear Programming formulation is proposed by drawing parallels with the Vehicle Routing Problem with Time Windows. The formulation employs a continuous-time representation to model the production sequence, while using discrete-time representations for inventory levels associated with each SKU and for quantities delivered to customers. Additionally, the paper proposes a diverse set of generated instances that reflect real-world industrial settings and uses them to validate the formulation through extensive computational experiments. The formulation is further strengthened by applying the Reformulation-Linearization Technique (RLT) to enhance the solvability of large-scale instances. The RLT-strengthened formulation demonstrates superior performance, yielding a significant improvement in the average optimality gap, and consistently produces solutions with an optimality gap of 2% in under an hour. The model is subsequently extended to incorporate dependencies between SKU productions, accompanied by extensive experimentation. Finally, comprehensive robustness analyses on key parameters are presented to provide managerial insights, identifying critical parameter ratios and thresholds for effective production scheduling.

3. **Minimum Triangular Inequalities**
(Co-authored with Qi Xiaofei, and Jitamitra Desai.)
Abstract: Most global optimization approaches for nonconvex QCQPs are based on constructing a convex or linear relaxation and embedding it with exhaustive search mechanism. Lower bound is computed as the value of objective function of this relaxation in each iteration of the search tree. In this paper, we propose a new class of valid inequalities to enhance the existing linearization (RLT-based) techniques to solve 0-1 QCQPs which we refer as Minimum Triangular Inequalities(MINTI). We prove that these new valid inequalities are tighter than the existing inequalities leading to improvement in the lower bound. This logic of MINTI cuts is extended to construct a novel variable fixing and branching strategy to identify feasible solution early in the search process which we refer as Minimum Triangle Algorithm(MITRA). Our computational results show the reduction in the number of nodes to attain optimality and in overall MITRA algorithm performs exceedingly well than existing methods.

TEACHING EXPERIENCE

Course Instructor

- *Probability and Statistics (Pre-Doctoral course)* Oct - Nov 2025
- *Introduction to Optimization (FOM course for undergraduates)* Apr 2025
  - Overall rating: 4.8/5 (No. of respondents: 36)
- *Introduction to MATLAB (PhD Preparatory Course)* Jun 2024
  - Overall rating: 4.58/5 (No. of respondents: 36)
- *Quantitative Technqiues (PGP Preparatory Course)* Jun 2024
  - Overall rating: 4.41/5 (No. of respondents: 25)

Teaching Assistant

- *Decision Sciences 2 (PGP)* Sep - Dec 2025
- *Optimization (PGPBA)* Jun - Sep 2025
- *Linear Programming and Networks (PhD)* Sep - Dec 2024
- *Decision Sciences 2 (PGP)* Sep - Dec 2024
- *Decision Sciences 1 (PGP)* June - Sep 2024
- *Optimization (PGPBA)* Jan - Mar 2024
- *Decision Sciences 2 (PGP)* Sep - Dec 2023
- *Decision Sciences 1 (PGP)* June - Sep 2023

CONFERENCE PRESENTATIONS

- **IISE Annual Expo & Conference 2026, Arlington, Dallas** May 2026
  A Continuous Time Model For Flexible Batch Scheduling With Sequence-Specific Changeovers.
- **MDC 2026, IIT Kharagpur** Jan 2026
  A Continuous Time Model For Flexible Batch Scheduling With Sequence-Specific Changeovers.
- **POMS India 2025, IIM Sambalpur** Dec 2025
  A Continuous Time Model For Flexible Batch Scheduling With Sequence-Specific Changeovers.
- **EURO 2025, University of Leeds** Jun 2025
  A Hierarchy of SOCP Relaxations for 0-1 Integer Programming Problems.
- **IISE Annual Expo & Conference 2025, Atlanta** May 2025
  A Hierarchy of SOCP Relaxations for 0-1 Integer Programming Problems.
- **MIP international 2024, IIT Bombay** Dec 2024
  A Hierarchy of SOCP Relaxations for 0-1 Integer Programming Problems.

WORKSHOPS

- Doctoral Consortium on Teaching, IIM Bangalore Jan 2025
- Mixed Integer Non-Linear Optimization, IIT Bombay Jan 2024

SERVICE & AFFILIATIONS

- **Indian Institute of Management, Bangalore**
  - Anti Ragging Committee, Jun 2024 - Jun 2025
  - PhD Hostel and Mess Representative Nov 2021 - Oct 2022
- **Indian Institute of Technology, Kharagpur**
  - Food & Mess Monitoring Committee, Dec 2018 - Jun 2019

SKILLS

MATLAB, GAMS, R, LaTeX, Microsoft Office, JMP, Mathematica

REFERENCES

Prof. Jitamitra Desai
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Prof. Arjun Ramachandra
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